

Project Title

AI-Driven Flood Prediction System for Urban Communities Using Google Earth Engine and ArcGIS – India

Non-Profit Organization/NGO Name: Assam State Disaster Management Authority (ASDMA)

Geographical Focus: Primary: Assam

Secondary: Bihar, Kerala, Telangana

- Assam State Disaster Management Authority (ASDMA) (Floods occur almost every year due to the Brahmaputra River)
- Bihar State Disaster Management Authority (Frequent flooding from the Kosi River)
- Kerala State Disaster Management Authority
- Greater Hyderabad Municipal Corporation (Urban flooding (Hyderabad floods 2020))

We want to include multiple flood-prone regions if our project is positioned as a scalable national solution. The system includes flood prediction models are data-driven, scalable and satellite data (GEE) works across regions.

Additional Partners: N/A

Number of Students Involved: 6

Number of Female Students: 3

Estimated Beneficiaries

- Short-term (1 year): 500–1000 people
- Long-term (3+ years): 5000+ people

Requested Funds (USD)

\$750

EPICS in IEEE Project Budget

Project Information

Project Title: AI-Driven Flood Prediction System for Urban Communities Using GEE & ArcGIS

IEEE Section: IEEE Hyderabad Section

University: Nalla Malla Reddy Engineering College

Project Lead: Dr. Sneha G

Financial Contact: 7892053937

Category	Item	Unit Cost (USD)	#of Units	Total Cost (USD)	Justification	Product Web URL
Physical Prototype / Software	Cloud Computing (Google Cloud/AWS credits)	100	2	200	Required for satellite data processing and ML model execution	https://cloud.google.com
	ArcGIS License / GIS Tools	150	1	150	GIS visualization and dashboard development	https://www.arcgis.com
Data Handling	Data Storage & Processing	50	2	100	Storage of satellite data, preprocessing datasets	N/A
Travel (Local only)	Field Visits for Validation	50	4	200	Travel for flood-prone area verification and NGO coordination	N/A
Logistics	Awareness Materials (posters, pamphlets)	20	5	100	Community awareness and training materials	N/A
Event Costs	Community Training / Workshop	0	0	0	Not included (optional if needed)	N/A

Project Details

Problem Statement

Floods caused by heavy rainfall and inadequate drainage systems are very common in urban settlements in monsoon affected areas. Floods cause traffic jams and affect infrastructure and people's safety. Conventional flood warning systems are dependent on expensive hardware devices such as sensors and cannot be adapted to changing conditions due to their lack of scalability and real-time processing ability. Therefore, there is a need for an efficient real-time flood prediction system.

Project Description

In this project, we plan to design an AI-based flood prediction system that will combine Google Earth Engine (GEE), ArcGIS, and Random Forest algorithms to identify floods. The proposed system will analyse satellite-derived environmental parameters like precipitation, elevation, slope, soil moisture, and vegetation indexes to detect potential flooding sites.

The model will classify locations as either low, medium, or high-risk areas and then display findings on GIS-based maps. The system will be flexible, affordable, and able to make real-time predictions.

Project Inputs

- **Software:** Google Earth Engine, ArcGIS, Python (Scikit-learn, NumPy, GDAL)

- **Data:** Satellite imagery, rainfall data, DEM, NDVI, soil moisture
- **Hardware:** Standard computers (minimum 8GB RAM)
- **Human Resources:** Students, faculty mentor, NGO collaboration
- **Cloud Resources:** Google Cloud / AWS

Project Activities and Outputs

1. Data Collection using Google Earth Engine

Data related to satellite and environment will be acquired through Google Earth Engine (GEE). The data that will be acquired includes rainfall intensity, soil moisture content, land surface temperature, Normalized Difference Vegetation Index (NDVI), Digital Elevation Model (DEM), slope, and land use land cover (LULC). The acquisition of the data will be through freely available sources such as Sentinel, Landsat, and global climate data. The data acquired will be on a time series basis.

2. Data Preprocessing and Feature Extraction

Preprocessing will include steps like removing noise, dealing with missing values, normalizing, and aligning the raster images in space. The datasets will be normalized to have a common resolution (e.g., 30 meters). Features will be extracted from these datasets to identify variables that influence flooding like drainage density, slopes, and vegetation index.

3. Model Development using Random Forest

Development of a Random Forest classification algorithm will take place using Python software (Scikit-learn). This model will be trained by the use of labelled data containing flood and non-flood classes. Hyperparameter tuning will be carried out for better results. The model captures non-linear relationships.

4. Model Testing and Validation

The data will be divided into a training set (70%) and a testing set (30%). The performance of the model will be measured by indicators like accuracy, precision, recall, F1 score, and ROC-AUC. Techniques for cross-validation will be used to enhance reliability. Moreover, the predicted flooded areas will be compared against the history of floods.

5. GIS-Based Visualization using ArcGIS

The outputs from the model will be analysed through the use of ArcGIS to create flood susceptibility maps. The raster layers will be classified based on three classes of risk – low, moderate, and high-risk areas. The maps will have visualization through colour coding – green for low, yellow for moderate, and red for high.

6. Deployment Planning (Dashboard/Alerts)

A deployment framework will be developed that ensures the system can be utilized by all users. It involves creating a web-based dashboard for GIS that helps monitor flood risks in real time. The system will also incorporate alerting facilities such as sending SMS or alerts on mobile devices to high-risk areas.

Outputs

1. Flood Vulnerability Map

This GIS map indicates vulnerable areas that have been classified according to low, medium, and high risk for flooding. This map will be used by decision-makers to determine which areas require attention.

2. Machine Learning Model

A machine learning algorithm (random forest classifier) trained for classifying areas into categories indicating vulnerability to floods depending on the environmental parameters.

3. GIS Flood Visualization Dashboard

A real-time dashboard showing flooding predictions and risk maps to help the users (authorities/NGOs) make decisions.

4. Risk Classification Reports

Reports presenting analyses of the results obtained, including predictions of floods, their risks, the efficiency of the model used, and the importance of features utilized in the prediction.

Engineering Design Process

The project follows a structured engineering design approach:

1. Problem Definition – Urban flooding problems
2. Requirement Definition – Real-time and scalable solution
3. Design – Incorporation of GEE, machine learning, and GIS
4. Implementation – Development of model and data pipeline
5. Testing – Validation with historical datasets
6. Evaluation – Evaluation based on accuracy, F1-score
7. Iteration – Model refinement and optimization
8. Deployment Planning – Integration with real-world systems

Project Maintenance and Sustainability

Cloud infrastructure shall be used in setting up the system, making it highly scalable and accessible. The system will be maintained by partner institutions like NGOs and local government bodies. Regular updating by use of satellite imagery shall be conducted to make sure that the information provided remains accurate throughout.

Project Impact

This project is intended to enhance disaster preparedness and minimize flood risks.

Impact Metrics

- Prediction accuracy
- Timely response rate
- User adoption rate
- Geographical area covered

Impact Scope

Short term: Enhanced awareness and early warning

Long term: Minimized economic impact

Student Learning Outcomes

Technical Skills

- AI Techniques (Random Forest)
- Geospatial Analysis (GEE, ArcGIS)
- Data Processing & Visualization

Soft Skills

- Collaborative teamwork
- Analytical thinking
- Community outreach

Students develop practical skills through the application of AI techniques to address societal issues.

Project Innovation

It is innovative because it incorporates the use of geospatial satellite data along with machine learning in real time to predict floods. In contrast to the use of sensors to detect floods, there is no need for setting up expensive hardware. When compared with deep learning methods, Random Forest gives good accuracy at low costs.

Deployment Strategy

The deployment strategy involves utilizing the application as a GIS dashboard which works seamlessly with ArcGIS Online. The system can send flood warnings through the use of mobile apps and SMS messages. The cloud computing platform guarantees that real-time information is accessible.

Project Budget

Category	Item	Cost (USD)
Cloud Resources	Storage & processing	\$200
Software/Tools	GIS tools (if required)	\$150

Data Handling	Storage & preprocessing	\$100
Travel	Field validation	\$200
Logistics	Awareness materials	\$100

Total Requested

\$750

Budget Justification

The proposed budget will be utilized for critical processes related to data processing, storage and field validation. Cloud computing enables scalability whereas funds are set aside for travel purposes.

Collaborators

NGO Details

Primary: ASDMA

Supporting: SEEDS India

Alignment with NGO Mission

The project is aligned with the NGO's mission of disaster risk reduction and community safety, offering an innovative flood prediction system that can be scaled and used in real-time.

Conclusion

The AI-based flood prediction system offers an efficient, cost-effective, and real-time approach to managing disasters. With the integration of machine learning algorithms and geospatial technology, this project not only improves early warning systems but also builds community resilience.